SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Blueprint

Autonomous Smart Caddy Cart (ASCC)

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Executive Summary

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Acronyms

AoA angle of arrival
ASCC Autonomous Smart Caddy Cart
BMS battery management system
CEP circular error probable
DC direct current
DC-DC direct current to direct current
FOC field-oriented control
GPIO general-purpose input/output
GPS Global Positioning System
HF high frequency
I2C Inter-Integrated Circuit
IP ingress protection
LiDAR light detection and ranging
MVP minimum viable product
NFC near-field communication
NMEA National Marine Electronics Association
RFID radio-frequency identification
RTK real-time kinematic
SPI Serial Peripheral Interface
UART universal asynchronous receiver-transmitter
UHF ultra-high frequency
UID unique identifier
UWB ultra-wideband

1. Introduction

1.1 Motivation and Problem

Golfers who walk the course must transport their equipment by carrying a bag or operating a push cart. Although electric and autonomous carts reduce this effort, they primarily focus on transportation rather than equipment management. Golfers must also keep track of clubs removed during play, which can be misplaced or left behind.

The Autonomous Smart Caddy Cart (ASCC) aims to address both challenges by combining autonomous golfer following with club identification and tracking. As golfers ourselves, the team understands the need for a solution that addresses both transportation and equipment management. The goal is to reduce manual cart operation while helping golfers manage their equipment throughout a round.

1.2 Project Objectives

The ASCC will be developed as a motorized golf push cart capable of following a golfer using a controller-based tracking system rather than computer vision. Distance sensors will support obstacle detection and avoidance, while an emergency-stop mechanism will allow the cart to be stopped when necessary.

An RFID-based subsystem will identify tagged golf clubs and monitor their presence in the bag, notifying the golfer when a club is missing. The objective is to integrate these functions into a reliable and safe prototype.

The ASCC will also aim to incorporate golf course data, including hole layouts, distances, and course information, to assist golfers with shot planning. This information helps golfers estimate distances to targets and hazards, supporting more informed club selection and on-course decisions.

1.3 Project Scope

The minimum viable prototype will focus on motor control, golfer following, obstacle detection, emergency stopping, RFID-based club tracking, and basic golf course data integration. Additional features, including shot data collection, club recommendations, and speech functionalities, may be implemented if time and resources permit.

Testing will begin indoors before progressing to controlled outdoor environments. The prototype is not intended for commercial deployment or operation on public roads, crowded areas, or extreme terrain.

1.4 System Overview and Operating Scenario

Figure 1 illustrates the ASCC's main subsystems and their interactions with the golfer and surrounding environment. The Raspberry Pi coordinates sensor inputs, motor control, and club tracking.

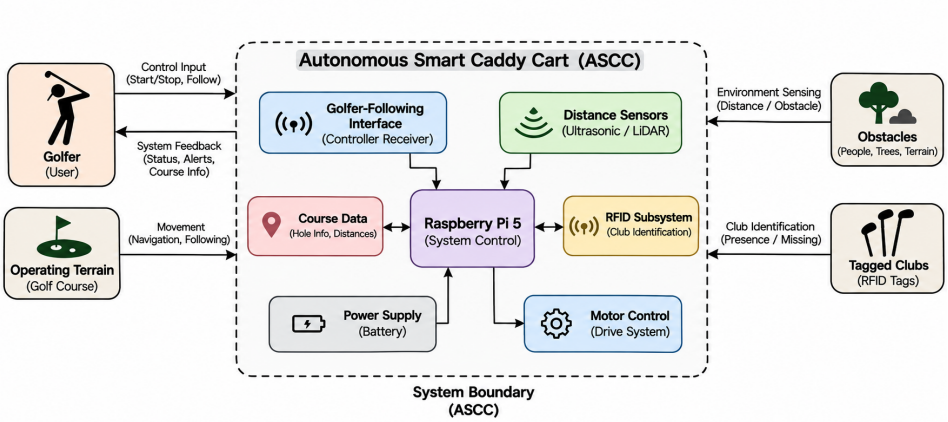


Figure 1: ASCC system overview and external interactions.

1.5 Design Constraints

The ASCC must be completed within the CAD $600 budget and capstone development schedule. Existing components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, will be reused to reduce costs.

Key constraints include hardware compatibility, sensor reliability, battery capacity, and safe motor operation. These considerations will guide the system requirements, development approach, and verification procedures.

2. Functional Description

2.1 System Overview and Design Alternatives

The Autonomous Smart Caddy Cart (ASCC) converts stored battery energy into motion toward one tagged golfer and decides, from radio, acoustic, and optical returns, whether that motion may continue. Club tracking and the distance display share the cart's power and processor but sit outside its motion path. The system boundary in Figure 1 includes the golfer's ultra-wideband (UWB) tag and the handheld e-stop. The golfer, safety operator, clubs, obstacles, satellites, and terrain are external. The main decisions are a used hoverboard with reflashed firmware as the drive base, a UWB tag for tracking the golfer, a stop path that bypasses the Pi, ultrasonic sensors plus light detection and ranging (LiDAR) for obstacles, passive radio-frequency identification (RFID) tags for clubs, and Global Positioning System (GPS) on the cart for distance. Table 19 (Appendix D) justifies each one against its alternatives and the needs that drive it.

2.2 Functional Decomposition

Figure 2 decomposes the system into six first-level functions: F1 move, F2 locate the golfer, F3 stop safely, F4 track clubs, F5 show distance, and F6 supply power. Each box performs one conversion of energy or information, and the labelled flows only carry its inputs and outputs. The separation of F3.4 from F3.7 and F3.8 is the central result. F3.4 runs on the Pi and brakes through I18, whereas F3.7 and F3.8 act on the energy path directly, so a hung Pi loses the brake but not the power cut. Table 20 (Appendix D) specifies the flows that cross a hardware boundary and traces them to the interface requirements.

2.3 Logical Decomposition and Interfaces

Figure 4 (Appendix D) allocates each function in Figure 2 to one logical element and states the action between elements. The following controller, safety controller, club inventory logic, and display composition are separate logical elements but share one physical processor, so a Pi fault is a common cause across them. This is why the safety interlock must act on the drive unit without passing through the safety controller, and why the mainboard enforces its own limits.

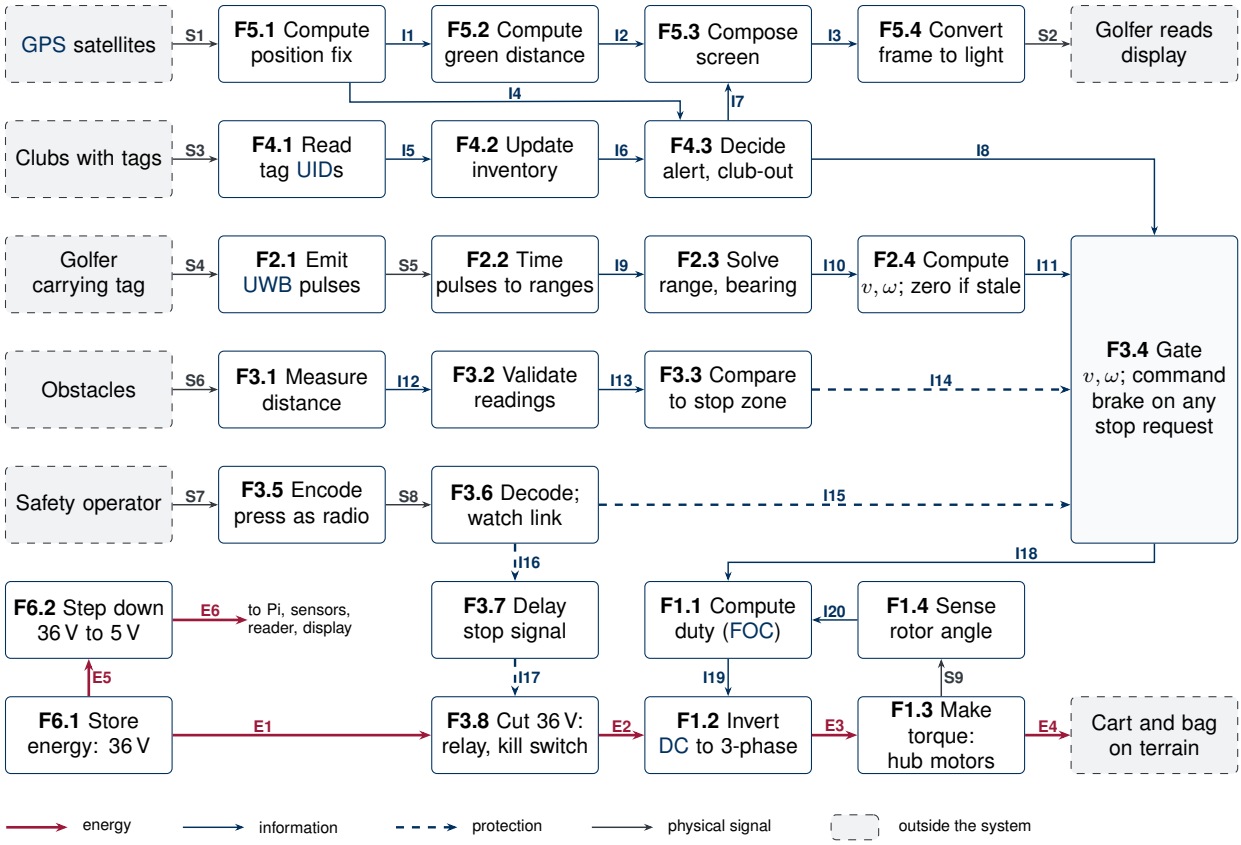


Figure 2: Functional decomposition. Labels E (energy), I (information), and S (physical signal) identify interfaces; Table 20 (Appendix D) traces them to requirements.

2.4 Physical Architecture

Figure 5 (Appendix D) places the components on the chassis. The hoverboard's two halves are locked at the pivot to form a rigid drive axle, with a front caster added. The pack and mainboard stay low to limit tip-over (H-05), and the obstacle sensors sit low at the front to see low objects (H-02). The UWB anchors are as far apart as the cross-bar allows, because bearing error scales with $1/b$. The GPS antenna is high on the handle and the sensitive antennas are kept away from the hub motors (C-4, C-5). Interfaces between the Pi and the rest of the cart use the buses in Table 20, and the compatibility checks at INT-1 confirm they do not conflict.

2.5 Prototype Definitions

The prototypes are P-01 drive base, P-02 tracking and following, P-03 safety layer, P-04 smart bag, P-05 distance display, and P-06 integrated ASCC. Table 21 (Appendix D) links each to its functions, physical elements, and tests, and Table 1 gives its development stages.

3. Requirements Specifications

3.1 Stakeholder Needs and Requirements Hierarchy

The ASCC requirements were developed around five stakeholder needs: equipment transportation, safety, club management, course information, and reliable operation. These address limitations of conventional motorized golf carts, which still require user control and do not assist with tracking equipment. The needs were decomposed into system-level and subsystem requirements to ensure each technical function contributes to the overall project objectives. Safety and essential movement functions were prioritized due to the risks associated with autonomous operation and the project's limited development resources. The requirements hierarchy is provided in Tables 4 and 5.

3.2 Functional Requirements

The primary function is autonomous golfer following, eliminating the need for continuous manual control. UWB tracking was selected to estimate the golfer's relative position without the computational demands and lighting dependencies of computer vision. Obstacle detection and emergency stopping provide safeguards when following commands could result in unsafe movement. RFID club tracking addresses the risk of forgotten equipment, while GPS-based course information improves convenience by displaying relevant distances and hole information. These functions were separated into independently testable requirements to support incremental development and integration, as detailed in Table 6.

3.3 Non-Functional and Interface Requirements

Performance targets were selected to balance usability, safety, and prototype feasibility. The 1–3 m following distance keeps the cart accessible while providing separation from the golfer. A target speed of 2 m/s allows the cart to keep pace with brisk walking, although normal operating speeds may be lower to maintain stability and reduce stopping distance. The initial 1 m obstacle detection range provides a measurable sensor benchmark, but the final stopping zone must account for braking distance, control latency, and sensor uncertainty.

The 90% RFID accuracy target establishes an initial reliability benchmark while accounting for tag orientation and reader placement. A 1-hour battery runtime supports extended testing without frequent recharging. These preliminary targets will be validated experi-

mentally, as outlined in Table 7. Interface requirements define communication and power exchange between subsystems, enabling independent development and reliable integration (Table 8).

3.4 Safety and Fault Requirements

Safety requirements prioritize preventing uncontrolled movement during emergency stops, tracking loss, and sensor faults. Emergency stopping must override autonomous commands and require manual reset to prevent unexpected restarting. A preliminary 1 s tracking timeout limits movement based on outdated position information, while invalid obstacle measurements must result in a safe stopped state rather than continued operation. These thresholds will be evaluated through braking tests and fault injection before outdoor testing. Tables 9 and 12 provide the detailed requirements and verification procedures.

3.5 Critical Requirements and Traceability

Requirements were classified as critical, major, or minor based on their importance to safety and prototype functionality. Critical requirements, particularly motor inhibition and obstacle response, must be verified before integrated autonomous testing. Traceability links each requirement to its stakeholder need, subsystem, prototype, and verification test, ensuring design decisions can be evaluated and changes tracked. For example, MC-04 requires motor actuation to be inhibited during an emergency stop and is verified through VT-07. The complete relationships are documented in Tables 10, 11, and 13.

3.6 Stretch Goals

Stretch goals include shot tracking, club recommendations, and voice-based interaction. These features could improve convenience and provide additional golfing insights but depend on reliable core subsystems. Voice recognition also introduces challenges from outdoor noise and cannot replace the dedicated emergency-stop mechanism. These features are therefore excluded from mandatory prototype acceptance criteria and will only be pursued if time and resources permit (Table 14).

4. Risks and Safety Hazards

4.1 Hazard Identification and Risk Assessment

Hazards were identified with the ISO 12100 process [1] and rated with the MIL-STD-882E risk matrix [2], where severity runs from 1 (catastrophic) to 4 (negligible) and probability from A (frequent) to E (improbable). Table 15 in Appendix B lists each hazard with its cause, governing standard, size, and rating before and after mitigation. The highest initial risk is uncommanded motion (H-03, serious), and all but one hazard remain medium after mitigation. Technical and schedule risks are covered as contingencies in Section 5.4.

The motion hazards depend mostly on speed. To stop within a gap $d_{\min}$ after a reaction time t_r, the cart must decelerate at

$$a_{\text{req}} = \frac{v^2}{2\,(d_{\min} - v\,t_r)}. \tag{1}$$

With $t_r = 0.3\,\text{s}$, stopping from 2 m/s (MC-03) within 1 m (FL-03) needs 5 m/s^2, which the cart's hub motors are unlikely to reach on grass, while walking pace (1.4 m/s) needs 1.7 m/s^2. The firmware speed cap is therefore the first safety measure, and DRV-4 checks the actual braking against MC-06 (C-9).

For H-09, a golfer only removes a club to swing it, and the swing reaches roughly arm length (about 0.7 m) plus club length (up to 1.22 m [3]), so a cart at the following distance can be inside it.

4.2 Risk Mitigation and Residual Risk

Table 16 and Figure 3 in Appendix B give the mitigation, derived requirement, and verification for each hazard. The key design choice is where the stops live. Speed and current limits and the serial-timeout stop run in the mainboard firmware, so they hold if the Pi crashes. The e-stop is a Category 1 stop [4]: it commands the firmware brake, then a relay cuts motor power without going through the Pi, since cutting power alone lets the hub motors freewheel. The cart does not restart until manually reset (MC-05) [5].

The remaining medium risks are accepted because a safety operator holds the e-stop during every moving test, on flat ground in dry weather. Those that become limitations in Section 8 are rollaway on a slope after power is cut (H-05), very thin or low objects the

sensors miss (H-02), a battery fire, which stays catastrophic in severity [6] (H-07), and a golfer swinging near a parked cart (H-09).

5. Approach

5.1 Development Strategy and Division of Labour

The ASCC will be developed incrementally, with each subsystem tested as its own prototype (P-01 to P-05) before it joins the full cart (P-06). The cart is driven by the hub motors, mainboard, and battery from a used hoverboard, with the mainboard reflashed to take speed commands from the Raspberry Pi.

The drive base (P-01), safety layer (P-03), and following (P-02) are built in that order, since following cannot be tested until the cart can drive and stop. The smart bag (P-04) and display (P-05) need no motion and are developed in parallel on the bench. The riskiest items are tested first, in November and December (DRV-1, BAG-1, TRK-1, TRK-2), and the e-stop (SAF-3) is verified before the first following test.

Jp leads P-01 and P-02, Connor leads P-04 and P-05, and Vince leads P-03 and the integration hardware. The whole team completes integration, and each member has milestones in every month (Table 17, Appendix C).

5.2 Prototype Development Plan

Table 1 defines each prototype. A milestone is complete when the matching stage of its test in Table 12 passes. P-03 includes a club-out interlock because a swing can reach the cart (H-09), and the smart bag already knows when a club is out.

5.3 Milestones and Dependencies

Table 17 (Appendix C) schedules the milestones from their dependencies and part lead times. Because Kingston grass is likely snow-covered from December to March, the grass test (DRV-4) runs on 27 November with a weight in place of the bag, and following and integration move indoors until March. Two weeks of buffer sit before INT-3 and one week before the final demonstration.

Integration is staged so the cart can always be stopped before anything is added: the motion chain first, then the bag and display on the stationary cart (INT-1), then the full system indoors (INT-2), then the field (INT-3, INT-4). INT-1 also checks power draw from

Table 1: Prototype definitions. Each stage ends at the milestone in brackets; Table 17 gives dates, leads, and dependencies.

Prototype, purpose, and scope	Development stages	Requirements	Tests
P-01 Drive base. Speed-limited motion on Pi commands, stoppable without the Pi. Reflashed mainboard, firmware limits, serial-timeout stop, kill switch, chassis	Bench, wheels raised (DRV-1) → limits and stops (DRV-2) → floor and grass with ballast (DRV-4) → chassis (DRV-3)	MC-01, MC-03, MC-06, IR-03	VT-01
P-02 Tracking and following. Follows the golfer's UWB tag, so it cannot switch to another person. Tag, two anchors or an AoA module, following controller	Static range (TRK-1) and bearing (TRK-2) → straight-line following (TRK-3) → turns, pace changes, second person (TRK-4)	FL-01 to FL-04, IR-01	VT-02, VT-08
P-03 Safety layer. Stops for obstacles, the e-stop, or a club out of the bag. Ultrasonic sensors, LiDAR, wireless e-stop, club-out interlock	Static detection (SAF-1) → e-stop, wheels raised (SAF-3) → stops while moving (SAF-2) → interlock (SAF-4)	OB-01 to OB-06, IR-02, MC-02, MC-04, MC-05, MC-07	VT-03, VT-04, VT-07, VT-11, VT-13
P-04 Smart bag. Tracks which clubs are in the bag and warns of one left behind. RFID reader, club tags, inventory software	One club in the bag (BAG-1) → full bag, stationary (BAG-2) → warning on cart movement (BAG-3)	CT-01 to CT-03, IR-04	VT-05
P-05 Distance display. Shows the distance to the green. GPS module, stored green coordinates, display	Surveyed points (GPS-1) → reference distances (GPS-2) → full display (GPS-3)	CD-01 to CD-03, IR-05	VT-06
P-06 Integrated ASCC. All subsystems on one cart, powered from the 36 V pack through a DC-DC converter	Stationary (INT-1) → indoors (INT-2) → field verification (INT-3) → golfer validation (INT-4)	PW-01, IR-06, all MVP requirements	VT-09, VT-10, all

the 36 V pack, Pi pin conflicts, motor noise on the RFID, UWB, and GPS readings, and loop timing (VT-10). Stretch goals (Table 18) start only after the MVP milestones they depend on have passed.

5.4 Contingency Plans

Table 25 in Appendix C lists the technical, safety, and schedule contingencies, each with its trigger, the change to the system or plan, and the milestones affected. The ones most likely to move the schedule are an unsupported mainboard (C-1), poor RFID reads near metal shafts (C-2), and early snow (C-6), which is why DRV-1, BAG-1, and DRV-4 are all due in November.

6. Verification and Validation

6.1 Verification Strategy and Traceability

Test T-X closes milestone X, and each objective VT-01 to VT-11 in Table 12 is carried out by the tests that name it in Table 2 and in Tables 22 and 23 (Appendix E). Test order follows the hazards: T-DRV-2 passes before the cart moves on its own wheels, and T-SAF-3 passes before the first following trial, because during following the operator is away from the cart. Every *shall not* requirement has a fault-injection step: MC-04 and MC-05 at T-SAF-3, FL-04 at T-TRK-3, and OB-04 and OB-05 at T-SAF-2. Hazard mitigations with no moving behaviour are verified by inspection (T-INS-1 to T-INS-3) and tip-over by analysis at T-DRV-3 [7].

Stop functions must pass 30 of 30 trials. Zero failures in 30 bounds the failure probability below 10% at 95% confidence ($0.9^{30} = 0.04$). That is the most a test campaign of this size can show, and it is why a safety operator stays present for all motion. Bench timing uses an oscilloscope in the ’65 ECE lab, field timing uses 240 fps video (about 4 ms per frame), and speed comes from mainboard Hall logs checked against video.

6.2 Subsystem and Prototype Tests

Table 2 gives the tests that verify the critical stop functions, using the set-ups in Figure 6 (Appendix E). Tables 22 and 23 in the same appendix give the rest, so that every prototype has its own tests. A red value has no Section 3 requirement yet.

T-DRV-2 tests configured values because the firmware defaults do not suit this cart. The default limit of 1000 rpm [10] is about 8.6 m/s on a 165 mm wheel [MEAS: wheel diameter], and the default timeout of 160 main-loop cycles at about 6 ms is near 1 s, or 2 m of travel at 2 m/s. T-DRV-4 also supplies values that later tests depend on. Measured deceleration and reaction time give the stopping distance $v\,t_r + v^2/2a$, which sets the validated stopping zone in OB-02 and OB-04, with OB-03 serving as the sensor benchmark. Deceleration below the value Eq. (1) requires at the speed cap triggers C-9.

6.3 Experimental Test Setups

Figure 6 (Appendix E) shows the arrangements behind the safety and tracking tests. The stop lane uses soft objects only, as required by H-02, and the bench setup measures

Table 2: Most critical verification tests. A failed criterion triggers the contingency listed in Table 25.

Test	Verifies	Set-up and procedure	Acceptance criteria
T-DRV-4 (P-01)	VT-01; MC-03; H-01, H-05	Figure 6(a), floor then short grass, ballast of [MEAS: loaded mass]. Accelerate to the cap, command zero speed, measure distance by video. Hold on [REQ: max slope]	$\geq$ 2.0 m/s; deceleration meets Eq. (1) at the cap, [REQ: braking]; no roll on the slope. Fail: C-7, C-9
T-TRK-3 (P-02)	VT-02, VT-08; FL-01 to FL-04	Indoors, straight lane, golfer at 1.4 m/s [8]; log range. Fault: switch off or shield the tag during a run, 30 runs	Range within 1–3 m for $\geq$ 95% of samples; speed command zero within 1 s of the last valid range, 30 of 30
T-SAF-2 (P-03)	VT-04, VT-11; OB-02, OB-04, OB-05	Figure 6(a) at the cap. Foam cylinders sized after the ISO 3691-4 test pieces [9], 200 $\times$ 600 mm and 70 $\times$ 400 mm, upright and lying. Fault: disconnect each sensor; inject out-of-range readings	Stops before contact, 30 of 30; detection to brake frame $\leq$ [REQ: 300 ms]; will not start with a sensor missing or invalid, 10 of 10 per sensor
T-SAF-3 (P-03)	VT-07; MC-02, MC-04, MC-05; H-04	Figure 6(b), then on the moving cart. Press at [REQ: 20 m]; repeat with the Pi halted and with the transmitter off; reset	Press to brake frame $\leq$ [REQ: 100 ms]; relay opens with the Pi halted; link loss stops within [REQ: 0.5 s]; no motion after reset until re-enabled; each 30 of 30. Fail: C-10
T-SAF-4 (P-03)	H-09; [REQ: add ID]	Following armed, floor. Remove a club and walk toward and away from the cart; log wheel commands; return the club	No motion toward the golfer while a club is out, 30 of 30; following resumes only after return

each stage of the stop separately, so a late stop can be traced to the radio, the Pi, or the mainboard.

6.4 System Validation

The verification tests check the requirements; T-INT-4 checks the stakeholder needs in intended use. A golfer plays a simulated hole of three shots on open, dry grass using only the cart, with no phone, three times on one charge and with the safety operator present. Table 24 (Appendix E) lists the observations for each need. If snow prevents this, C-6 applies and the deviation is stated.

7. Resource Requirements

The ASCC has a project budget of CAD $600. Several key components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, are already owned or borrowed, reducing the overall cost. The remaining budget provides flexibility for unexpected ex-

penses, replacements, or design changes. Additional components, including motors and GPS tracking hardware, may be obtained through the ECE Department or Ingenuity Labs. However, their estimated costs are included in the budget until their availability is confirmed. Table 3 summarizes the estimated costs.

7.1 Hardware and Equipment

Table 3: Estimated ASCC bill of materials (CAD).

Component	Qty.	Cost
Raspberry Pi 5 (existing)	1	$0
Raspberry Pi display	1	$40
Ultrasonic sensors (existing)	Multiple	$0
RFID readers and tags	2 sets	$20
TF-Luna LiDAR sensor	1	$50
GPS tracker	1	$30
Golf push cart (existing)	1	$0
Motors	1 set	$80
Motor controller	1	$50
Battery and power supply	1 set	$75
Mounting hardware	Assorted	$30
Wiring, connectors, switches	Assorted	$30
Tax and shipping	–	$75
Total		**$480**

7.2 Facilities and Contingencies

Development will require electronics tools and access to indoor and outdoor testing areas. Tools can be found utilizing the ’65 ECE lab, and testing will occur in controlled areas and hopefully on a real golf course for the finished prototype. Python and Git will support software development.

The estimated cost is $480, leaving $120 below the $600 budget limit. This remaining budget allows for additional components or design adjustments if needed.

8. Conclusion

The ASCC blueprint expands upon the initial proposal by establishing a structured system design, measurable requirements, safety analysis, and detailed development plan. The project aims to integrate golfer following, obstacle detection, emergency stopping, RFID club tracking, and GPS-based course information into a functional autonomous golf cart.

Stakeholder needs have been decomposed into functional, performance, interface, and safety requirements, each linked to planned verification tests. Quantitative targets for following distance, speed, obstacle detection, RFID accuracy, and battery runtime provide measurable acceptance criteria. The hazard analysis further identifies risks involving collisions, uncontrolled movement, electrical faults, and terrain conditions, with corresponding safeguards and residual risks documented.

Development will follow an incremental approach, prioritizing motor control, safety, and tracking before full system integration. Critical functions will be verified through subsystem testing and controlled fault injection before outdoor validation. Technical failures are addressed through defined contingency plans, including alternative tracking methods, RFID configurations, hardware replacements, and adjustments to operating limits. Remaining limitations include sensor blind spots, battery hazards, and restricted terrain suitability.

With an estimated cost of CAD $480 against the $600 budget, the project remains financially feasible. The next steps are to finalize quantitative safety thresholds, confirm component compatibility, and complete the initial hardware proofs of concept. Successful completion will be demonstrated through verified subsystem requirements and an integrated prototype capable of supporting a simulated golf hole under controlled conditions.

9. References

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A. Detailed Requirements and Traceability

This appendix provides the complete ASCC requirements hierarchy, subsystem specifications, and verification traceability. Requirements are organized from stakeholder needs (SN) to system-level requirements (SYS) and subsystem requirements. Each requirement is assigned a priority and linked to a planned verification test.

A.1 Stakeholder Needs and System Requirements

Table 4: Stakeholder needs and associated system requirements.

ID	Stakeholder Need	System Requirements
SN-01	Transport golf equipment without manually pushing the cart.	SYS-01
SN-02	Prevent collisions and unintended movement.	SYS-02
SN-03	Identify clubs and notify users of missing equipment.	SYS-04
SN-04	Access course information and distances.	SYS-05
SN-05	Maintain reliable operation within project constraints.	SYS-06

Table 5: System-level requirements and classifications.

ID	System Requirement	Priority
SYS-01	The ASCC shall autonomously follow a golfer carrying the designated tracking controller.	Major
SYS-02	The ASCC shall prevent forward movement into detected obstacles within its stopping zone.	Critical
SYS-03	The ASCC shall provide an emergency-stop function that overrides autonomous movement commands.	Critical
SYS-04	The ASCC shall monitor the presence of individually tagged golf clubs.	Major
SYS-05	The ASCC shall provide available golf course information through a user interface.	Minor
SYS-06	The ASCC shall support battery-powered operation during controlled testing.	Major

A.2 Functional Subsystem Requirements

Table 6: Detailed functional subsystem requirements.

ID	Requirement	Parent
FL-01	The following subsystem shall calculate the required movement direction from valid tracking-controller information.	SYS-01
FL-02	The following subsystem shall update its movement command when the golfer changes position relative to the cart.	SYS-01
OB-01	The obstacle subsystem shall measure distances to objects in the forward detection region.	SYS-02
OB-02	The obstacle subsystem shall issue a stop request when a detected obstacle enters the stopping zone.	SYS-02
MC-01	The motor-control subsystem shall execute valid movement commands received from the main controller.	SYS-01
MC-02	The motor-control subsystem shall disable motor actuation in response to an emergency-stop request.	SYS-03
CT-01	The club-tracking subsystem shall associate detected RFID identifiers with registered golf clubs.	SYS-04
CT-02	The club-tracking subsystem shall generate a missing-club alert when a registered club remains absent beyond the configured detection interval.	SYS-04
CD-01	The course-data subsystem shall display available hole information for the selected course.	SYS-05
CD-02	The course-data subsystem shall display an estimated distance to a selected course target when valid location data is available.	SYS-05

A.3 Performance and Operational Requirements

Table 7: Detailed performance and operational requirements.

ID	Requirement	Parent
FL-03	The ASCC shall maintain a following distance between 1 and 3 m during controlled walking trials.	SYS-01
MC-03	The ASCC shall achieve a forward speed of at least 2 m/s on level terrain under the specified test load.	SYS-01
OB-03	The obstacle subsystem shall detect the designated test obstacle at a distance of at least 1 m.	SYS-02
CT-03	The RFID subsystem shall correctly identify at least 90% of registered clubs during repeated detection trials.	SYS-04
PW-01	The power subsystem shall support at least 1 hourof continuous operation under representative test conditions.	SYS-06
MC-06	The ASCC shall decelerate at no less than 2 m/s^2 on level grass under the specified test load.	SYS-02
CD-03	The course-data subsystem shall display the distance to the green within 5 m of a surveyed reference distance.	SYS-05

A.4 Interface Requirements

Table 8: Detailed interface requirements.

ID	Requirement	Parent
IR-01	The following interface shall transmit tracking information to the main controller.	SYS-01
IR-02	The obstacle sensors shall transmit distance measurements to the main controller.	SYS-02
IR-03	The main controller shall transmit movement commands to the motor-control subsystem.	SYS-01
IR-04	The RFID interface shall transmit detected tag identifiers to the main controller.	SYS-04
IR-05	The user interface shall display club-status information generated by the club-tracking subsystem.	SYS-04
IR-06	The power subsystem shall provide the required operating voltages to the computing and motor-control electronics.	SYS-06

A.5 Safety and Fault Requirements

Table 9: Detailed safety and fault requirements.

ID	Requirement	Parent
MC-04	The ASCC shall not energize its drive motors while the emergency-stop control is engaged.	SYS-03
MC-05	The ASCC shall not resume autonomous movement following an emergency stop until a manual reset is performed.	SYS-03
FL-04	The ASCC shall not continue autonomous movement when the tracking signal has been unavailable for more than 1 s.	SYS-03
OB-04	The ASCC shall not continue forward movement when an obstacle is within the validated stopping zone.	SYS-02
OB-05	The ASCC shall not initiate autonomous movement when required obstacle sensor data is unavailable or invalid.	SYS-02
OB-06	The ASCC shall not move toward the golfer while a registered club is absent from the bag.	SYS-02
MC-07	The ASCC shall begin braking within 100 ms of an emergency-stop command or loss of the emergency-stop link.	SYS-03

A.6 Complete Requirements Traceability Matrix

Table 10: Functional and performance requirements traceability.

ID	Parent	Need	Priority	Prototype	Test
FL-01	SYS-01	SN-01	Major	P-02	VT-02
FL-02	SYS-01	SN-01	Major	P-02	VT-02
FL-03	SYS-01	SN-01	Major	P-02	VT-02
FL-04	SYS-03	SN-02	Critical	P-02	VT-08
OB-01	SYS-02	SN-02	Critical	P-03	VT-03
OB-02	SYS-02	SN-02	Critical	P-03	VT-04
OB-03	SYS-02	SN-02	Critical	P-03	VT-03
OB-04	SYS-02	SN-02	Critical	P-03	VT-04
OB-05	SYS-02	SN-02	Critical	P-03	VT-11
OB-06	SYS-02	SN-02	Critical	P-03	VT-13
MC-01	SYS-01	SN-01	Major	P-01	VT-01
MC-02	SYS-03	SN-02	Critical	P-01	VT-07
MC-03	SYS-01	SN-01	Major	P-01	VT-01
MC-04	SYS-03	SN-02	Critical	P-01	VT-07
MC-05	SYS-03	SN-02	Critical	P-01	VT-07
MC-06	SYS-02	SN-02	Critical	P-01	VT-12
MC-07	SYS-03	SN-02	Critical	P-01	VT-07
CT-01	SYS-04	SN-03	Major	P-04	VT-05
CT-02	SYS-04	SN-03	Major	P-04	VT-05
CT-03	SYS-04	SN-03	Major	P-04	VT-05
CD-01	SYS-05	SN-04	Minor	P-05	VT-06
CD-02	SYS-05	SN-04	Minor	P-05	VT-06
CD-03	SYS-05	SN-04	Minor	P-05	VT-06
PW-01	SYS-06	SN-05	Major	P-06	VT-09

Table 11: Interface requirements traceability.

ID	Parent	Need	Priority	Prototype	Test
IR-01	SYS-01	SN-01	Major	P-02	VT-10
IR-02	SYS-02	SN-02	Critical	P-03	VT-10
IR-03	SYS-01	SN-01	Major	P-01	VT-10
IR-04	SYS-04	SN-03	Major	P-04	VT-10
IR-05	SYS-04	SN-03	Major	P-05	VT-10
IR-06	SYS-06	SN-05	Major	P-06	VT-10

A.7 Verification Test Definitions

Table 12: Planned verification tests.

ID	Test	Verification Objective
VT-01	Motor movement	Verify movement commands and target speed.
VT-02	Golfer following	Measure following distance and response to golfer movement.
VT-03	Obstacle detection	Measure detection range and sensor response.
VT-04	Obstacle response	Verify stopping behaviour when obstacles enter the stopping zone.
VT-05	RFID identification	Measure identification accuracy and missing-club alerts.
VT-06	Course information	Verify hole information and displayed distances.
VT-07	Emergency stop	Verify motor inhibition and manual-reset behaviour.
VT-08	Signal loss	Verify stopping behaviour following tracking-signal loss.
VT-09	Battery runtime	Measure continuous operating duration.
VT-10	Interface integration	Verify information and power exchange between subsystems.
VT-11	Sensor fault	Verify safe response to missing or invalid obstacle measurements.
VT-12	Braking	Measure deceleration and stopping distance on level grass under the test load.
VT-13	Club-out interlock	Verify the cart does not move toward the golfer while a club is out of the bag.

A.8 Critical Requirement Relationships

Table 13: Critical requirement relationships.

ID	Function	Logical Element	Verification
MC-04	Inhibit motor actuation	Safety interlock and motor-control subsystem	VT-07
FL-04	Detect loss of tracking	Following controller and signal monitor	VT-08
OB-04	Prevent forward movement near obstacles	Obstacle monitor and safety controller	VT-04
OB-05	Detect invalid sensor information	Sensor validation logic	VT-11

A.9 Stretch Goal Requirements

Table 14: Optional stretch goal requirements.

ID	Optional Requirement
SG-01	The ASCC may record shot locations and distances for later review.
SG-02	The ASCC may recommend golf clubs using course information and recorded shot data.
SG-03	The ASCC may provide speech-based status updates and user interaction.
SG-04	The ASCC may provide a voice functionality to provide information through sound.
SG-05	The ASCC may provide a voice speech recognition feature for commands.

B. Detailed Hazard Analysis

Table 15 lists the hazards assessed in Section 4.1. Table 16 gives the mitigations for each one by ISO 12100 step, the requirement each produces, and the test or milestone that verifies it. Figure 3 shows how these controls act on the main hazard, loss of motion control.

Table 15: Safety hazards on the ASCC, the standard that governs each one, the size of the risk, and the risk level before and after mitigation.

ID	Hazard and cause	Standard	Size of the risk	Before	After
H-01	**Cart hits the golfer.** The golfer stops suddenly, or walks back to the bag for a club and ends up closer than the following distance	ISO 13482 [11], ISO 3691-4 [9]	Stopping from 2 m/s within 1 m needs about 5 m/s^2 of braking (Eq. 1); about 60 J at impact	3C Medium	3E Medium
H-02	**Cart hits something it did not detect.** The ultrasonic beams can miss low objects such as sprinkler heads and curbs, and grass can cause false readings	ISO 13482	Objects must be detected beyond the stopping distance; OB-03 requires 1 m. Blind zone below [MEAS: sensor height]	3C Medium	3D Medium
H-03	**Cart moves without a valid command.** The UWB tag is dropped or handed to someone else, a UART command is corrupted, the Pi crashes, or the firmware speed limit is set wrong	ISO 13482	Up to 2 m of travel at 2 m/s during the 1 s allowed by FL-04; uncapped speed [MEAS: stock top speed]	2C Serious	3E Medium
H-04	**E-stop fails or is too slow.** The radio link drops, the stop is routed through a frozen Pi, or the cart restarts after a reset	ISO 13850 [5], IEC 60204-1 [4]	0.2 m of extra travel for every 100 ms of latency at 2 m/s	2D Medium	2E Medium
H-05	**Cart rolls away or tips over.** Power is cut on a slope and the hub motors freewheel, or the cart turns across a slope with a tall, loaded bag	ISO 13482	Tips at $\arctan(w/2h)$ for track width [MEAS: w] and centre of gravity [MEAS: h], compared with the site slope	3C Medium	3D Medium
H-06	**Fingers or straps caught.** The exposed hub wheels sit close to the frame and caster	ISO 13857 [12], ISO 13854 [13]	Measured gaps [MEAS: gaps] compared with the minimum gaps for fingers and hands	3D Medium	4D Low
H-07	**Battery fire, burn, or shock.** The 36 V lithium-ion pack, new wiring to the DC-DC converter, short circuits, and charging	UL 2272 [14], IEC 62133-2 [15]	About 158 Wh stored in a typical 36 V, 4.4 Ah pack [MEAS: pack label]	1E Medium	1E Medium
H-08	**Water reaches the electronics.** Dew on the grass or rain during an outdoor test	IEC 60529 [16]	Enclosure IP rating compared with [REQ: IP rating]	3D Medium	3E Medium
H-09	**Golfer's swing and the cart.** The golfer swings at a cart within reach, or the cart moves into the golfer mid-swing	Equipment Rules [3]	A swing reaches about 1.9 m, which covers most of the 1–3 m following distance in FL-03	3C Medium	3E Medium

Table 16: Mitigations, derived requirements, and verification for each hazard.

ID	Mitigations	Derived requirement	Verification
H-01	*Design:* speed limit in the firmware, and a minimum following distance chosen using Eq. (1). *Safeguard:* the cart does not move toward the golfer inside that distance, and the firmware brakes it to a stop	Shall not move toward the golfer when closer than 1 m (FL-03); shall decelerate at 2 m/s^2 or more (MC-06)	VT-02, VT-12; DRV-4, TRK-3
H-02	*Design:* LiDAR and downward-angled ultrasonic sensors that cover low objects. *Safeguard:* obstacle stop. *Information:* only soft obstacles are used in testing	Shall not move forward with an obstacle in the stopping zone (OB-04); shall detect the test obstacle at 1 m (OB-03)	VT-03, VT-04; SAF-1, SAF-2
H-03	*Design:* speed and current limits on the mainboard, independent of the Pi. *Safeguard:* serial-timeout stop, tag-loss timeout, rejection of sudden jumps in range, and a watchdog on the Pi	Shall not keep moving more than 1 s after tracking is lost (FL-04); shall not exceed [REQ: speed cap] under any command	VT-01, VT-08; DRV-2, TRK-3
H-04	*Safeguard:* Category 1 stop (Section 4.2), with link loss treated as a stop, and a manually pressed button held by the safety operator	Shall not resume motion until manually reset (MC-05); shall begin braking within 100 ms (MC-07)	VT-07; SAF-3
H-05	*Design:* low bag cradle and the widest track the chassis allows. *Safeguard:* the firmware holds the wheels still while powered. *Information:* no operation on steep slopes	Shall not be operated on slopes above [REQ: slope limit]	DRV-4 (hold on a slope); tip calculation
H-06	*Design:* guards over the hub wheels. *Information:* switch off the kill switch before any adjustment	Guard gaps shall meet ISO 13857	T-INS-1 (inspection)
H-07	*Design:* stock pack and BMS left unmodified, inline fuse sized to the wiring, and strain relief. *Information:* charge with the original charger, off the cart, while supervised	The accessory circuit shall be fused at [REQ: fuse rating]	T-INS-2 (inspection); current measured at INT-1
H-08	*Design:* rated enclosure. *Information:* dry weather only, and testing stops if it rains	The enclosure shall meet [REQ: IP rating]	T-INS-3 (inspection)
H-09	*Safeguard:* club-out interlock that blocks motion toward the golfer until the club is returned. *Information:* park the cart outside swing reach	Shall not move toward the golfer while a club is out of the bag (OB-06)	VT-13; SAF-4

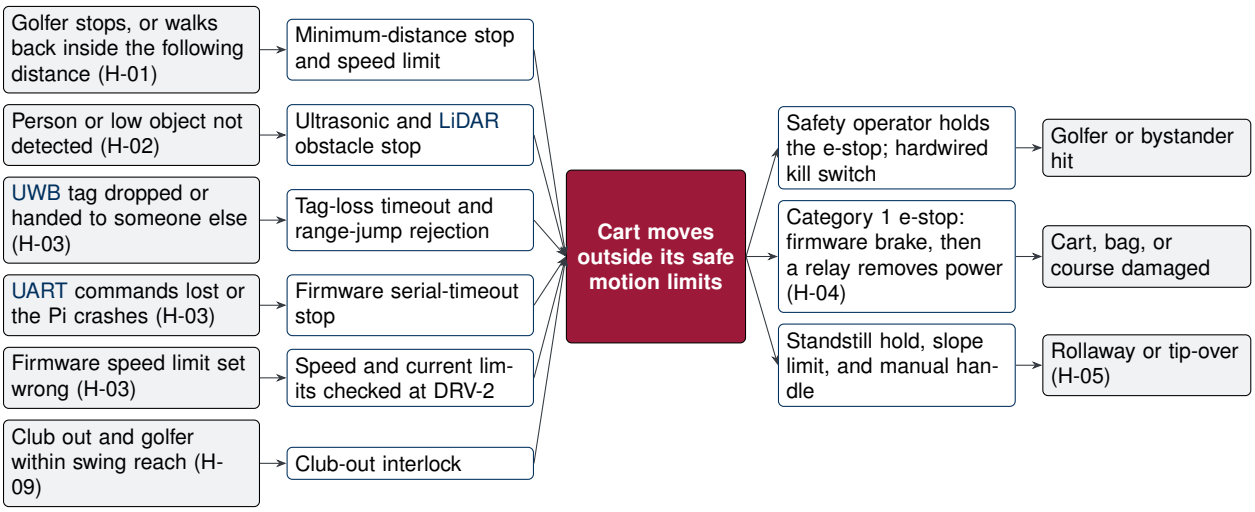


Figure 3: Bow-tie diagram for loss of motion control, showing causes and preventive controls on the left and protective controls and consequences on the right.

C. Milestone Schedule and Contingencies

Table 17 gives the dates, owners, dependencies, and verification tests for each stage in Table 1. A milestone is complete when the matching stage of its test in Table 12 passes. Table 18 lists the stretch milestones, which map to the stretch goals in Table 14.

Table 17: MVP milestones. All due dates are Fridays between November 2026 and March 2027.

ID	Milestone	Depends on	Lead	Due	Test
P-01 Drive base					
DRV-1	Firmware flashed, and the Pi spins both wheels over UART with the wheels raised	–	Jp	13 Nov	VT-01
DRV-2	Speed and current limits, serial-timeout stop, and kill switch checked on the bench	DRV-1	Jp	20 Nov	VT-01
DRV-4	Speed and braking measured on the floor and on grass, with a weight equal to a loaded bag	DRV-2	Jp	27 Nov	VT-01, VT-12
DRV-3	Chassis finished, with the hub motors, bag cradle, handle, caster, and enclosure mounted	DRV-1	Jp	4 Dec	–
P-03 Safety layer					
SAF-1	Obstacles detected with the cart static, including low objects	–	Vince	27 Nov	VT-03
SAF-3	Wireless e-stop stops the cart on a press or when its link drops, and needs a manual reset	DRV-2	Vince	11 Dec	VT-07
SAF-2	Cart stops for obstacles and for faulty sensor data while moving	DRV-3, DRV-4, SAF-1	Vince	22 Jan	VT-04, VT-11
SAF-4	Cart does not move toward the golfer while a club is out of the bag	BAG-2, TRK-3, INT-1	Vince	19 Feb	VT-13
P-04 Smart bag					
BAG-1	One tagged club read reliably inside the real bag, near metal shafts	–	Connor	27 Nov	VT-05
BAG-2	Every club in a full, stationary bag reported as present or absent	BAG-1	Connor	11 Dec	VT-05
BAG-3	Warning given when the cart moves away without a removed club	BAG-2, GPS-1	Connor	29 Jan	VT-05
P-05 Distance display					
GPS-1	Position error measured at surveyed outdoor points	–	Connor	27 Nov	VT-06
GPS-2	Distance to the green compared with reference distances	GPS-1	Connor	15 Jan	VT-06
GPS-3	Display shows the distance, club inventory, and warnings	GPS-2, BAG-2	Connor	5 Feb	VT-06
P-02 Tracking and following					
TRK-1	UWB range to the golfer's tag measured while static	–	Jp	11 Dec	VT-02
TRK-2	UWB bearing to the golfer's tag measured while static	TRK-1	Jp	15 Jan	VT-02
TRK-3	Straight-line following indoors, stopping when tracking is lost	DRV-3, DRV-4, TRK-2, SAF-3	Jp	29 Jan	VT-02, VT-08
TRK-4	Following through turns and changes of pace, without switching to a second person	TRK-3	Jp	12 Feb	VT-02
P-06 Integrated ASCC					
INT-1	Bag and display mounted on the stationary cart and powered from its battery, with compatibility checks passed	DRV-3, BAG-3, GPS-3	Vince, Connor	12 Feb	VT-10
INT-2	Full system running on flat indoor ground	INT-1, TRK-4, SAF-2, SAF-4	All	26 Feb	VT-10
INT-3	Field test of every MVP requirement, including battery runtime	INT-2	All	12 Mar	VT-01 to VT-11
INT-4	A golfer plays a simulated hole using only	INT-3	All	19 Mar	See 6

Table 18: Stretch milestones. Each one starts only after the milestones it depends on have passed.

ID	Stretch milestone	Goal	Depends on	Main difficulty
STR-1	Voice commands to stop, follow, and wait	SG-05	TRK-3	Wind noise outdoors causes missed commands. A spoken “stop” can only be an extra stop, because ISO 13850 expects the e-stop to be pressed by hand [5]
STR-2	Spoken distance and status updates	SG-03, SG-04	GPS-2, BAG-3	Needs a speaker on the cart that can be heard outdoors
STR-3	Shot distances logged from where each club was taken out	SG-01	BAG-2, GPS-1	The cart’s position is not the ball’s position, and club-tag trackers also miss shots [17]
STR-4	Club recommendations from logged distances	SG-02	STR-3	Needs enough logged shots with each club to be useful

D. Functional Description Details

Table 19 supports Section 2.1. Table 20 traces the interfaces labelled in Figure 2 to their requirements. Figures 4 and 5 support Sections 2.3 and 2.4, and Table 21 supports Section 2.5.

Table 19: Design decisions, their drivers, and the alternatives rejected.

Decision and drivers	Why, and why not the alternatives
DD-1 Used hoverboard, mainboard reflashed with open-source FOC firmware [10] *SN-01, SN-05; MC-01, MC-03; budget*	One unit supplies two Hall-sensed hub motors for differential steering, the inverter, and a 36 V pack with BMS. Closed-loop speed mode holds speed under load, and speed and current limits run on the mainboard rather than the Pi (H-03). Hoverboard parts have carried robots outdoors in public [18], on an ODrive controller rather than the stock board [19]. *Used motorized golf cart:* the proposal's $375 for cart, controller, and battery takes 63% of the budget. *Separate motors and controller:* adds the wheels, mounts, and control tuning that the hoverboard includes. Accepted cost: small wheels on grass (C-7).
DD-2 UWB tag on the golfer, anchors on the cart *SN-01; FL-01 to FL-04*	Ranges to one identified tag and is minimally affected by light [20]. Anchors on a robot reached 162 mm mean position error after correction (703 mm before) [21], and one AoA anchor on a stroller reached 15 cm and 5° at 2 m [20]. Both were indoors, so grass and a tag blocked by the golfer's body are tested at T-TRK-2. *Camera following* [22]: must re-identify the golfer when a second person is near, and competes with the stop loops for the Pi. *Bluetooth handset* [23]: signal-strength ranging suffers from multipath [24]. *GPS:* 2.5 m CEP [25] is as large as the 1–3 m following band. Two-anchor ranging or AoA is decided at T-TRK-2: error falls with anchor spacing [21], so ±10 cm ranging [26] over a 0.5 m baseline b gives about $\sqrt{2}\sigma_r/b \approx 16°$.
DD-3 Category 1 e-stop with a hardware path that bypasses the Pi *SN-02; SYS-03; MC-02, MC-05; H-03, H-04*	Section 4.2 sets the sequence of brake, then relay. *Firmware-only stop:* a hung Pi defeats it, and the firmware's own serial timeout only removes motor power, so the wheels coast [10]. *Category 0 stop:* hub motors freewheel. *Voice stop:* ISO 13850 expects a manually actuated device [5].
DD-4 Ultrasonic array plus TF-Luna LiDAR *SN-02; OB-01 to OB-05; H-02*	The MVP stops rather than steers, so it needs distance in the forward path, not a map. The TF-Luna ranges 0.2–8 m at 90% reflectivity and 0.2–2.5 m at 10%, but its 2° beam sees one point [27]; the existing ultrasonic sensors cover the cart's width and low objects. Combining sonar with LiDAR has detected obstacles such as glass that LiDAR alone misses [28], indoors and with a 360° LiDAR. *One sensor type:* narrow coverage, or false returns from grass. *Camera:* as in DD-2. *Scanning LiDAR:* free-space shape is needed only for STR-8.
DD-5 Passive RFID tags on clubs, reader in the bag *SN-03; CT-01 to CT-03*	Identifies each club with no battery or golfer input. *Per-slot presence sensors:* lose club identity (C-2). *Club-mounted trackers* [29]: need a phone and can miss shots [17]. Known weakness: nearby metal detunes tag antennas, as shown at UHF [30], and a low-cost HF reader typically operates only up to 50 mm [31], so tag position and antenna layout are set by T-BAG-1, not assumed.
DD-6 GPS on the cart, green coordinates stored *SN-04; CD-01, CD-02*	Receiver accuracy of 2.5 m CEP [25] and golf devices at about three yards [17] support a 5 m target. *Watch or phone* [32]: the dependence the ASCC removes. *RTK:* cost not justified at this accuracy. Accepted cost: distance is measured from the cart, not the ball.

Table 20: Interfaces crossing a hardware boundary, with the requirement each traces to.

Label	From → to	Specification	Req.
S5, I9	F2.1 → F2.3	UWB channel 5 (6.5 GHz) or 9 (8 GHz); ranges to the host over SPI [26]	IR-01
I12	F3.1 → F3.2	TF-Luna: UART at 115200 baud or I2C, 3.3 V logic, 100 Hz default [27]; ultrasonic trigger and echo on GPIO	IR-02
I15–I17	F3.6 → F3.4, F3.8	Fail-safe stop line: high is run; low, a broken wire, or link loss is stop. Reaches the relay without passing through the Pi	[REQ: add ID]
I18	F3.4 → F1.1	UART at 115200 baud, 3.3 V logic; the 15 V wire in the same cable stays unconnected [10]	IR-03
I5	F4.1 → F4.2	Tag UIDs over SPI, up to 10 Mbit/s [31]	IR-04
I1	F5.1 → F5.2	NMEA position over UART, at least 1 Hz [25]	[REQ: add ID]
I3, I7	F5.3, F4.3 → F5.4	Distance, inventory, and alerts on a display of at least 5 in diagonal	IR-05
E1, E2	F6.1 → F1.2	36 V nominal, 10-cell pack (42 V full); 15 A per motor default limit [10]	IR-06
E6	F6.2 → electronics	5 V; the Pi 5 alone draws up to 5 A [33]	IR-06

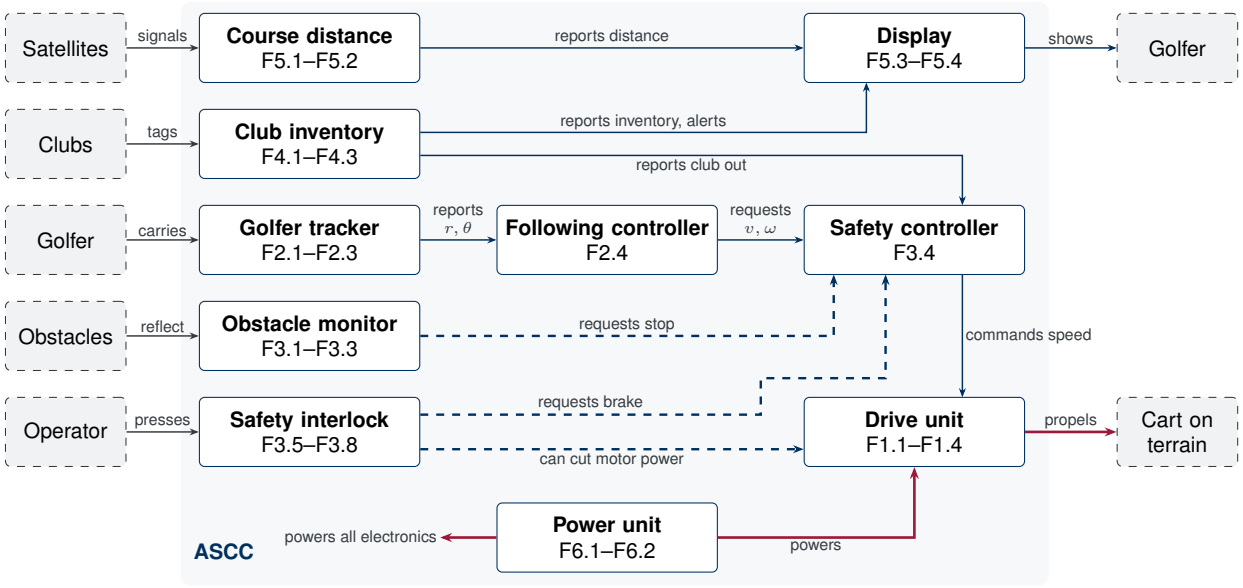


Figure 4: Logical decomposition, with allocated functions in each element and the action on each link. Line styles match Figure 2.

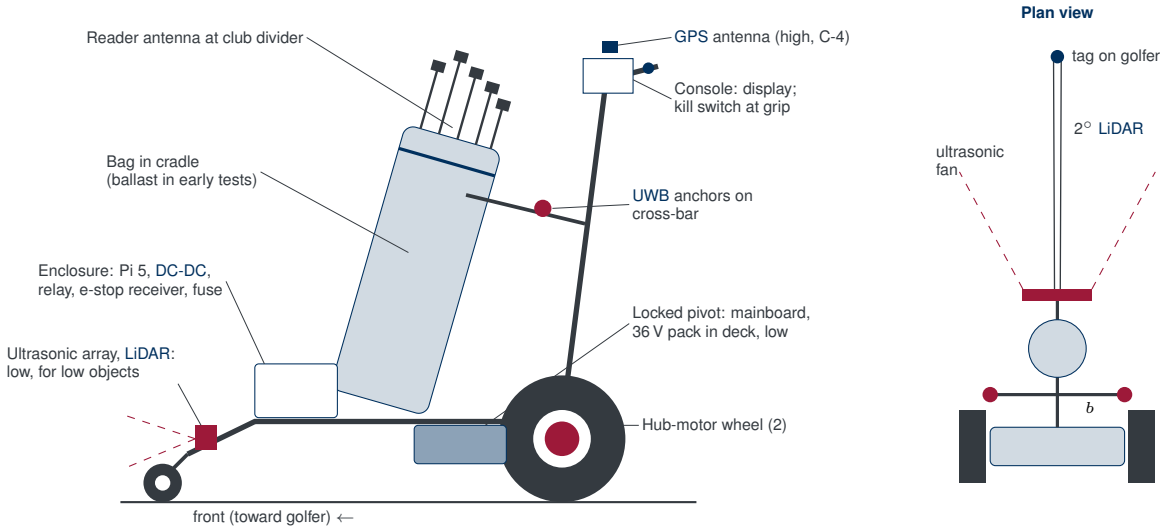


Figure 5: Physical layout, side elevation and plan view (not to scale). The tag and the handheld e-stop are carried off the cart.

Table 21: Prototypes, with the functions, physical elements, and tests for each.

ID, name	Functions	Physical elements	Tests
P-01 Drive base	F1, F6.1, kill switch	Hoverboard mainboard, hub motors, and pack; chassis, caster, handle	T-DRV-1 to 4, T-INS-1
P-02 Tracking and following	F2	UWB tag and anchors; following software on the Pi	T-TRK-1 to 4
P-03 Safety layer	F3	Ultrasonic array, TF-Luna; handheld e-stop, receiver, relay; safety software on the Pi	T-SAF-1 to 4
P-04 Smart bag	F4	RFID reader and club tags; inventory software on the Pi	T-BAG-1 to 3
P-05 Distance display	F5	GPS module, display; distance software on the Pi	T-GPS-1 to 3
P-06 Integrated ASCC	All, F6.2	P-01 to P-05 on one cart; DC-DC converter, enclosure, fuse	T-INT-1 to 4, T-INS-2, T-INS-3

E. Detailed Test Plan

Figure 6 shows the set-ups for Table 2 and the tests below. Table 22 gives the remaining key tests for each prototype, and Table 23 gives the interface, proof-of-concept, and hazard-mitigation tests they rely on. A red value has no Section 3 requirement yet. Table 24 gives the validation in Section 6.4.

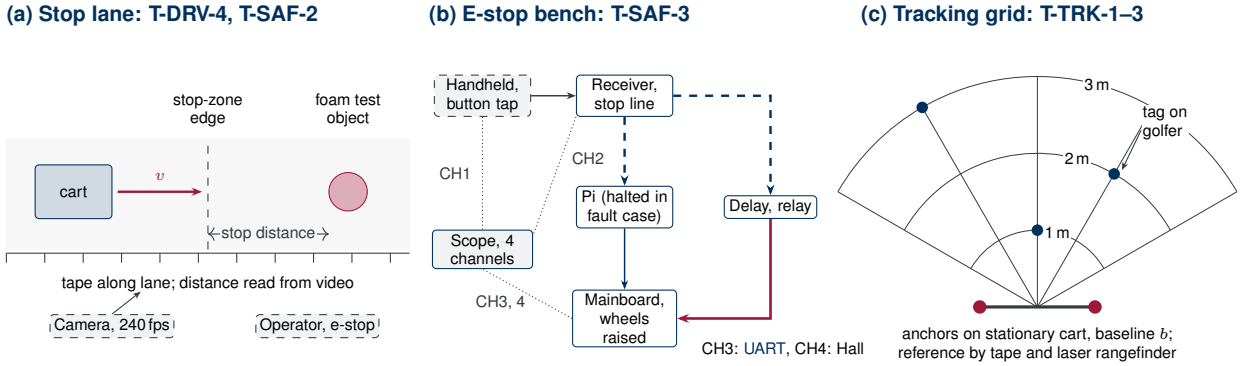


Figure 6: Experimental set-ups: (a) stop lane, (b) e-stop timing bench, (c) tracking grid.

Table 22: Other key verification tests. A failed criterion triggers the contingency listed in Table 25.

Test	Verifies	Set-up and procedure	Acceptance criteria
T-DRV-2 (P-01)	VT-01; MC-01; H-03	Wheels raised. Command twice the cap over UART and log speed; stall a wheel and log current; stop sending frames; pull the kill switch	Speed $\leq$ [REQ: cap: 2.2 m/s]; serial loss removes power within [REQ: 0.25 s]; kill switch removes power, 10 of 10
T-TRK-2 (P-02)	VT-02; FL-01	Figure 6(c), cart stationary. Tag at each grid point for 60 s, then again with the golfer's body between tag and anchors	Mean bearing error $\leq$ [REQ: 15°] at 1–3 m and $\pm$60°. Fail: C-3
T-BAG-2 (P-04)	VT-05; CT-01, CT-03	Full 14-club bag, stationary. Remove and return clubs singly and in groups, 5 cycles per club	$\geq$ 63 of 70 events correct (CT-03); no swapped identity; delay $\leq$ [REQ: 5 s]. Fail: C-2
T-GPS-2 (P-05)	VT-06; CD-02	Outdoors at reference points 50–200 m from stored coordinates, cart stationary. Compare with a laser rangefinder	Displayed distance within [REQ: 5 m] at every point. Fail: C-4
T-INT-3 (P-06)	VT-09; PW-01; all MVP	Short grass, loaded bag, from full charge. Follow on continuous laps with pack voltage logged; repeat T-SAF-2, T-SAF-3, and T-TRK-3 faults, 10 trials each, with all processes running	$\geq$ 1 h with no brownout (PW-01); all repeated trials pass. Fail: C-5, C-6

Table 23: Remaining verification tests. Set-ups match Figure 6 where they apply.

Test	Verifies	Set-up and procedure	Acceptance criteria
T-DRV-1 (P-01)	VT-01, VT-10; MC-01, IR-03	Wheels raised. Pi sends speed frames at three setpoints in both directions	Hall speed within $\pm$10% of command; no firmware fault. Fail: C-1
T-DRV-3 (P-01)	H-05	Inspection and analysis. Lock the pivot, load the cradle with ballast, measure track w and centre of gravity height h	No relative motion at the pivot; tip angle $\arctan(w/2h)$ exceeds [REQ: max slope]
T-TRK-1 (P-02)	VT-02, VT-10; IR-01	Figure 6(c), then an open area out to 15 m. Tag at each point for 60 s against tape	Range error $\leq$ [REQ: 0.3 m] at 1–3 m; valid ranges to [REQ: 13 m]. Fail: C-3
T-TRK-4 (P-02)	VT-02; FL-02, FL-03	Golfer turns $\pm$90° and varies pace; a second person with no tag crosses between golfer and cart, 10 runs	Range within 1–3 m for $\geq$ 90% of samples; cart never follows the second person
T-SAF-1 (P-03)	VT-03, VT-10; OB-01, OB-03, IR-02	Static. Place the test objects on a grid across the forward region, 0.2–3 m, 20 readings per point; map the blind zone	Detected at $\geq$ 1 m across the cart's width (OB-03); blind zone recorded for H-02
T-BAG-1 (P-04)	VT-05, VT-10; CT-01, IR-04	Proof of concept. One tagged club among metal shafts in the real bag; measure read range; 50 reads per tag position	$\geq$ 45 of 50 correct in the installed position. Fail: C-2
T-BAG-3 (P-04)	VT-05; CT-02	Move the cart away with one club out, 10 runs; then 10 runs with all clubs in	Alert within [REQ: 20 m] of movement, 10 of 10; no false alert in 10 runs
T-GPS-1 (P-05)	VT-06	Log fixes for 10 min at surveyed outdoor points	95% of fixes within [REQ: 5 m]. Fail: C-4
T-GPS-3 (P-05)	VT-06, VT-10; CD-01, IR-05	Daylight. A reader at 3 m, the upper limit of FL-03, reads the distance, inventory, and alerts	Correct in 10 of 10 readings; diagonal $\geq$ [REQ: 5 in]
T-INT-1 (P-06)	VT-10; IR-01 to IR-06	The four compatibility checks in Section 5.3, with the 5 V rail on an oscilloscope during motor acceleration and subsystem tests repeated with motors running	No undervoltage flag on the Pi; no bus conflict; T-BAG-2, T-TRK-1, and T-GPS-1 still pass; loop periods hold. Fail: C-5
T-INT-2 (P-06)	All MVP	Flat indoor course with following, an obstacle stop, an e-stop, and a club removal, all processes running	Correct response to every event; no watchdog reset
T-INS-1 (P-01)	H-06	Measure guard gaps with calipers	Gaps meet ISO 13857 [12]
T-INS-2 (P-06)	H-07	Inspect fuse, wiring, and strain relief; measure accessory current at INT-1	Fuse rating below wire ampacity and above measured current
T-INS-3 (P-06)	H-08	Inspect enclosure and cable entries	Meets [REQ: IP rating: IP54] [16]

Table 24: Validation of stakeholder needs at T-INT-4.

Stakeholder need	Scripted event or observation	Passes when
SN-01 Transport without pushing	Golfer walks between shots at their own pace	Golfer never touches the cart to move it
SN-02 No collisions or unintended movement	Soft obstacle placed in the path; one scripted e-stop	No contact; no unscripted operator stop; no motion toward the golfer with a club out
SN-03 Missing clubs	A club is deliberately left at the second shot	Display names that club before the third shot
SN-04 Course information	Golfer states the distance at each shot from the display only	Stated distance within 5 m of the laser reference
SN-05 Reliable operation	All three runs on one charge	No software restart or unscripted reset

Table 25: Contingency plans.

ID	Trigger	Change to the system or plan	Milestones
C-1	Mainboard unsupported, or flashing fails (DRV-1)	Use the community firmware fork for that board, or buy another used hoverboard for parts with a supported mainboard	DRV-1 up to 2 weeks late; DRV-4 nears snow (C-6)
C-2	Under 90% of clubs read near metal shafts (CT-03, BAG-1)	On-metal NFC tags with one antenna per slot on an SPI multiplexer. Last resort: per-slot presence sensing, which loses club identity (CT-01)	BAG-2 up to 2 weeks late; BAG-3, GPS-3, INT-1 follow
C-3	Bearing error too large to hold 1–3 m (FL-03, TRK-2)	Switch between two-anchor ranging and an AoA module	TRK-3, TRK-4, INT-2
C-4	Distance error above 5 m (CD-03, GPS-1)	Average readings while the cart is stopped, and add an active antenna on the handle	GPS-2
C-5	Readings fail or the Pi browns out with the motors running (INT-1)	Read the bag only when stopped, give the Pi its own DC-DC supply, add ferrites, and move antennas away from the motors	INT-1; INT-2 buffer absorbs 2 weeks
C-6	Snow before DRV-4, INT-3, or INT-4	Grass tests in November; INT-3 and INT-4 on a cleared lot or after snowmelt, reported as a deviation	DRV-4, INT-3, INT-4
C-7	Wheels slip or stall on grass (DRV-4)	Limit operation to short, flat grass, or switch to the larger hub motors from an off-road hoverboard	DRV-4, INT-3
C-8	A track falls 2 or more weeks behind	A member whose track is ahead joins it, and stretch work stops	INT-2, INT-3; buffer absorbs 2 weeks
C-9	Braking below 2 m/s^2 (MC-06, DRV-4)	Lower the speed cap and raise the minimum distance until Eq. (1) holds; revise MC-03 and FL-03	DRV-4; TRK-3 uses the new limits
C-10	E-stop slower than 100 ms (MC-07), or link drops (SAF-3)	Dedicated industrial e-stop receiver wired directly to the power relay	SAF-3; TRK-3 waits for it